

Year 12 (L6) — Homework & Consolidation Plan, 2026/27

Homework and consolidation tasks to **round out each week's new content** and **build confidence in previously-covered topics** through deliberate spaced retrieval and interleaving. Aligned to the L6 CMP Scheme of Work and designed for the **~4.5 hrs/week** of independent study (IST/VESPA).

How this is designed

- **Homework** = consolidates the **week's new content** (a printable quiz + a focused skills task + the programming strand).
- **Consolidation** = *rounds out the week and revisits earlier topics* on a spacing schedule (1 day → 1 week → 1 month → cumulative). This is the part that **builds confidence in prior topics** and protects against forgetting.
- Everything links to resources already in this repo, so tasks are zero-prep to set.

Resource shorthand (all in this repo)

Short name	What / where
Quiz x.y.z	Printable subtopic quiz → revision-tools/printable-pdfs/subtopic-quizzes/
Worksheet x.y	Printable worksheet → revision-tools/printable-pdfs/worksheets/
Flashcards x.y	Deck → revision-tools/flashcards/
KO x.y	Knowledge organiser → revision-tools/printable-pdfs/knowledge-organisers/
Mini-paper x.y	Timed topic mock → revision-tools/printable-pdfs/mini-papers/
Mock 1 / 1b	Full 140-mark mock → revision-tools/printable-pdfs/mock-papers/
Workbook 1 / 2	Programming workbook → revision-tools/programming-workbook/
NEA guide	Project guide → component-03-04-programming-project/
Guides	Glossary, reference sheet, common mistakes, tracker → revision-tools/guides/

Retrieval rule of thumb: every consolidation includes a short *closed-book* retrieval of a topic from earlier in the course. Keep a one-line **error log** of dropped marks and re-RAG the [self-assessment tracker](#).

Autumn Term 1 (Weeks 1–7)

Wk (w/b)	Topic	 Homework	 Consolidation (round-out + retrieval)
1 (31/08)	Induction & Diagnostic	Complete the diagnostic task (knowledge + research/programming); verify VS Code + Python + Google Classroom set up.	Read the subject booklet; build your notes & folder structure (VESPA <i>Systems</i>); skim the course README to see the year ahead.
2 (07/09)	1.1.1 Processor structure & function (CPU, FDE, performance, Von Neumann/Harvard)	Quiz 1.1.1 (closed-book, self-mark) + the register/FDE sections of Worksheet 1.1 . Programming: Workbook 1 §1–2 (sequence/selection/iteration) ★.	Make & learn Flashcards 1.1 (registers, FDE steps); cover-and-recall KO 1.1 . Retrieval: redo any diagnostic questions you missed.
3 (14/09)	1.1.2 Types of processor (pipelining, CISC/RISC, GPUs, multicore) · KA1	Revise for Key Assessment 1 with KO 1.1 + Flashcards 1.1 ; then Quiz 1.1.2 .	Interleave: re-do Quiz 1.1.1 Section A closed-book. Programming: Workbook 1 strings/lists ★.
4 (21/09)	1.1.3 Input/output & storage · 2.1.1 Thinking abstractly · KA1 feedback	Quiz 1.1.3 + Quiz 2.1.1 . Act on KA1 feedback — redo your two weakest questions.	Retrieval of 1.1.1–1.1.2 via Flashcards 1.1 (shuffled). Programming: Workbook 1 (operators).
5 (28/09)	1.2.1 Systems software (OS, scheduling, interrupts) · 2.1.2 Thinking ahead · CAP1/KA2	Quiz 1.2.1 (Sections A–B); start Quiz 2.1.2 ; prepare for CAP1/KA2 .	Spaced retrieval of 1.1: Mini-paper 1.1 Section A, timed. Programming: Workbook 1 (selection).
6 (05/10)	<i>Review week 1</i> · 1.2.1 (BIOS, drivers, virtual machines) · <i>CAP1/KA2 feedback</i>	Finish Quiz 1.2.1 ; act on CAP1/KA2 feedback (error log).	File-check prep — organise folder (VESPA <i>Systems</i>). Retrieval: mixed Flashcards 1.1 + 1.2 . Programming: Workbook 1 (iteration).
7 (12/10)	1.2.2 Applications generation (translators/compilers) · 2.1.2 Thinking ahead (preconditions, caching, reuse) · KA3 · <i>CAP1 due 12th</i>	Quiz 1.2.2 + complete Quiz 2.1.2 ; stages-of-compilation ordering drill (Worksheet 1.2).	Cumulative retrieval before half term : full Mini-paper 1.1 , timed & marked. Programming: Workbook 1 (nested structures).

Half term (19/10): optional low-stakes retrieval pack — finish any outstanding quiz, 10 min/day of **Flashcards 1.1–1.2**. No heavy homework.

Autumn Term 2 (Weeks 8–14)

Wk (w/b)	Topic	 Homework	 Consolidation (round-out + retrieval)
8 (02/11)	1.2.3 Software methodologies (+ algorithms & test strategies) · 2.1.3 Thinking procedurally · <i>KA3 feedback</i>	Quiz 1.2.3 + methodology-match task (Worksheet 1.2); Quiz 2.1.3 .	Spaced retrieval of 1.1 + 1.2.1/1.2.2 (mixed flashcards / quiz Section A). Programming: Workbook 1 (string manipulation).
9 (09/11)	1.2.4 Types of programming language (paradigms, assembly/LMC, addressing modes, OOP) · CAP2/KA4	Quiz 1.2.4 (incl. addressing-mode calculation); LMC practice (topic 1.2 worked example); revise for CAP2/KA4 with KO 1.2 .	Retrieval: Mini-paper 1.1 + Flashcards 1.2.1–1.2.3 . Programming: Workbook 1 (arrays).
10 (16/11)	1.3.1 Compression & encryption · 1.3.2 Databases intro · <i>CAP2/KA4 feedback</i>	Quiz 1.3.1 (incl. an RLE encode/decode); begin Quiz 1.3.2 ; act on CAP2 feedback .	Cumulative 1.1 + 1.2 retrieval — Mini-paper 1.2 , timed. Programming: Workbook 1 (lists & tuples).
11 (23/11)	1.3.2 Databases (normalisation, SQL) · <i>file checks</i>	Complete Quiz 1.3.2 ; SQL-writing tasks (Worksheet 1.3); 1NF → 3NF normalisation practice (topic 1.3 worked example).	Retrieval: cover-and-recall KO 1.2 . Programming: <i>Using Databases</i> — write SQL in code (Workbook).
12 (30/11)	1.3.3 Networks (referential integrity, TCP/IP, LAN/WAN, hardware, P2P/client-server) · <i>RAD day 1</i>	Quiz 1.3.3 (incl. TCP/IP layer matching); redraw the networks/TCP-IP diagrams from the Diagram bank (guides/).	Spaced retrieval 1.1–1.2 (mixed quiz). Programming: Workbook 1 (functions, input validation).
13 (07/12)	2.1.4 Thinking logically · 1.3.4 Web technologies (HTML/CSS, JavaScript) · <i>CAP2 due 7th</i>	Quiz 2.1.4 ; Quiz 1.3.4 Sections A–B (HTML/CSS/JS).	Retrieval of 1.3.1/1.3.2 (flashcards/quiz Section A). Programming: Workbook 1 (file handling).
14 (14/12)	1.3.4 Web technologies (server/client-side, indexing, PageRank) · <i>End-of-year quiz</i>	Finish Quiz 1.3.4 ; write a PageRank explanation (topic 1.3); prep the end-of-year quiz with KO 1.1–1.3 .	Christmas retrieval pack : mixed Flashcards 1.1–1.3 + one Mini-paper . Programming: Workbook 1 (OOP).

Christmas holidays: spaced retrieval pack — one **Mini-paper (1.1, 1.2 or 1.3)** every few days + 10 min/day flashcards. Keep it light and spaced.

Spring Term 1 (Weeks 15–20)

Wk (w/b)	Topic	 Homework	 Consolidation (round-out + retrieval)
15 (04/01)	1.4.1 Data types (positive/negative binary, add/subtract, hex)	Quiz 1.4.1 Section A + binary/hex conversion & two's-complement drills (Worksheet 1.4); keep the reference sheet to hand.	New-year cumulative retrieval 1.1–1.3 (mixed flashcards + a quiz Section A). Programming: Workbook 1 (file processing).
16 (11/01)	1.4.1 Data types (floating point, arithmetic, bitwise) · KA5	Floating-point normalisation drills (Worksheet 1.4 + topic worked example); revise for KA5 with the reference sheet .	Retrieval of 1.3 (quiz/flashcards). Re-do any binary questions you slipped on.
17 (18/01)	1.4.1 (character sets, data types) · 2.2.1 Programming techniques (constructs) · KA5 feedback · <i>U6 mock week</i>	Complete Quiz 1.4.1 ; Quiz 2.2.1 Sections A–B; act on KA5 feedback (error log).	Cumulative 1.1–1.3 — Mini-paper , timed. Programming: PyGame basics (Component 3 start).
18 (25/01)	1.4.2 Data structures (arrays, lists, records, tuples; linked lists; stacks/queues) · CAP3/KA6	Quiz 1.4.2 (incl. a stack/queue trace); revise for CAP3/KA6 with KO 1.4 .	Retrieval: 1.4.1 calc drills + Flashcards 1.2 . Programming: Workbook 2 — implement a Stack & Queue class.
19 (01/02)	<i>Review week 2</i> · 1.4.2 (implementing linked lists/stacks/queues — Component 3) · CAP3/KA6 feedback	Implement a linked list / stack / queue in Python (Workbook 2); file-check prep.	Retrieval of 1.3 — Mini-paper 1.3 . Programming: PyGame game states.
20 (08/02)	1.4.2 (graphs, trees, hash tables) · <i>RAD day 2</i> · CAP3 due 8th	Remainder of Quiz 1.4.2 ; implement a BST + hash table (Workbook 2).	Cumulative 1.1–1.4.1 mixed retrieval. Programming: PyGame custom events.

Half term (15/02): retrieval pack — **Flashcards 1.4** + one earlier **Mini-paper**; finish any data-structures implementation.

Spring Term 2 (Weeks 21–25)

Wk (w/b)	Topic	 Homework	 Consolidation (round-out + retrieval)
21 (22/02)	1.4.3 Boolean algebra (logic gates, truth tables) · KA7	Quiz 1.4.3 Section A + truth-table completion; revise for KA7 with KO 1.4 .	Retrieval of 1.4.1/1.4.2 (calc + structure traces).
22 (01/03)	Component 3 — PyGame (sprite classes, sounds, high scores) · <i>KA7 feedback</i>	Component 3 PyGame tasks (sprite classes); act on KA7 feedback .	Retrieval: Flashcards 1.4 + a Mini-paper .
23 (08/03)	1.4.3 Boolean logic (Karnaugh maps, simplifying) · <i>Recap 1.2</i>	Boolean simplification / De Morgan's / Karnaugh-map drills (Worksheet 1.4 + topic); complete Quiz 1.4.3 .	Recap 1.2: Mini-paper 1.2 , timed + KO 1.2 cover-recall.
24 (15/03)	CAP4/KA8 · 2.1.5 Thinking concurrently · <i>Recap 1.3</i> · <i>Review week 3</i>	Quiz 2.1.5 ; revise for CAP4/KA8 (cumulative 1.1–1.4 + 2.1).	Recap 1.3: Mini-paper 1.3 + Flashcards 1.3 .
25 (22/03)	<i>CAP4/KA8 feedback</i> · Component 3 PyGame · <i>Recap 1.4</i>	Component 3 PyGame workbook; act on CAP4/KA8 feedback .	Recap 1.4: Mini-paper 1.4 + reference-sheet number/logic drills.

Easter holidays: structured **mock prep** — work through **Mock Paper 1 (Computer Systems)** a section at a time across the break, self-mark against the scheme, and RAG-rate. Top up weak areas with the matching **subtopic quizzes**.

Summer Term 1 (Weeks 26–32) — Mocks & Mini-NEA

Wk (w/b)	Topic	 Homework	 Consolidation (round-out + retrieval)
26 (12/04)	Component 3 PyGame · <i>Mock exam revision</i>	Sit Mock Paper 1 under timed conditions; self-mark; RAG-rate every topic.	Target your weakest RAG topics with the matching subtopic quizzes / KOs ; continue PyGame.
27 (19/04)	L6 mock week (CAP5)	Final focused prep — Mock Paper 1b + error-log review (light; it's mock week).	Skim glossary + common-mistakes sheet (<i>guides/</i>); rest before exams.
28 (26/04)	Component 3 PyGame · <i>mock feedback</i>	PyGame workbook; act on mock feedback — re-revise your weakest topics.	Re-attempt your 2 weakest mock questions; flashcards on those topics.
29 (03/05)	Component 3 — Mini-NEA: Analysis · CAP5 due 7th	Analysis using the NEA guide : named stakeholder, requirements, measurable success criteria .	NEA-guide checklist — confirm every success criterion is <i>testable</i> . Maintain theory with 10 min/day flashcards.
30 (10/05)	Mini-NEA: Analysis	Complete the analysis write-up (stakeholder research + success criteria).	Peer-review a partner's analysis against the NEA marking checklist.
31 (17/05)	Mini-NEA: Design	Design: decomposition, algorithms (pseudocode/flowcharts), data structures, UI, test plan (NEA guide) .	Check each design decision links back to a requirement.
32 (24/05)	Mini-NEA: Design	Complete the design + a test plan with typical/boundary/erroneous data.	Review the design against the NEA design checklist.

Half term (31/05): finish the NEA design so implementation can start cleanly; light flashcard retrieval to keep theory warm.

Summer Term 2 (Weeks 33–36) — Mini-NEA build

Wk (w/b)	Topic	 Homework	 Consolidation (round-out + retrieval)
33 (07/06)	Mini-NEA: Implementation	Iterative development with annotated evidence and testing each iteration (NEA guide) ; reuse techniques from Workbook 1 & 2 .	Keep a development log ; test as you build (don't leave testing to the end).
34 (14/06)	Mini-NEA: Implementation · <i>Review week 4</i>	Continue implementation; add validation/error handling (robustness).	File-check prep — ensure all code evidence is annotated .
35 (21/06)	Mini-NEA: Evaluation	Evaluate against each success criterion with evidence; gather stakeholder feedback (NEA guide).	Map every evaluation point back to an analysis criterion.
36 (28/06)	Mini-NEA: Evaluation	Complete the evaluation + justified improvements .	Final NEA checklist; reflect on VESPA targets going into Year 13.

Notes for the teacher

- **Differentiation:** quizzes have Section A (recall) → C (exam-style); set Section A for all, B–C for stretch. Workbook tasks are tagged ★/★★/★★★.
- **Low-prep marking:** every quiz, worksheet and mini-paper has a **full answer key / mark scheme**, so they're self-markable or cover-teacher-friendly.
- **The confidence loop:** the *Consolidation* column is the spaced-retrieval spine — it revisits each topic at ~1 week, ~1 month and cumulatively, which is what moves earlier content into long-term memory and steadies confidence before each Key Assessment/CAP.
- **Adjust freely:** swap a mini-paper for flashcards if time is short; the principle (new content in Homework, prior content in Consolidation) is what matters.